

$^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$ 2016Mu03

$J^\pi=0^+$ for ^{36}S ground state.

2016Mu03: A secondary beam of ^{36}S was produced via the projectile fragmentation of a 140-MeV/nucleon ^{48}Ca primary beam impinging on an 846-mg/cm² ^9Be target at the coupled cyclotron facility at NSCL, MSU. The ^{36}S nuclei were selected using the A1900 separator with an intensity of 8.1×10^5 pps and a purity of 89.7%. The excited states of ^{35}P were populated by the one-proton knockout reaction from the ^{36}S beam at 88 MeV/nucleon (midtarget energy) on a 100-mg/cm² ^9Be secondary target located at the reaction target position of the S800 spectrometer. The projectile-like residues were identified from their energy loss measured by an ionization chamber located at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators situated at the object position and at the focal plane of the S800 spectrometer. Prompt γ rays from the deexcitation of the ^{35}P residues were detected by seven modules of the GRETINA Ge array. Measured E_γ , I_γ , $(^{35}\text{P})\gamma$ -coin, $\gamma\gamma$ -coin, the inclusive knockout cross section for producing ^{35}P from ^{36}S , the fractional populations and parallel momentum distributions of 8 populated states in ^{35}P residues. Deduced levels, J , π , L-transfers, partial knockout cross sections and spectroscopic factors. Calculated single-particle cross sections (σ_{sp}) for proton removal and parallel momentum distributions using the eikonal model. This work is also part of the thesis [2015MuZY](#).

 ^{35}P Levels

b_f values under comments are population fraction in the knock-out reaction, σ_f^{sp} values are for calculated single-particle cross sections, and R_s values are quenching factors from a systematics study in [2014To14](#). The partial cross sections to each level can be deduced from $b_f \times \sigma_{\text{inc}}^{\text{exp}}$.

The inclusive cross section $\sigma_{\text{inc}}^{\text{exp}}$ for producing ^{35}P from ^{36}S is measured to be 51 mb *I*. 95.3% is attributed to a direct knockout process from the p-sd shell, i.e., $\sigma_{\text{inc,KO}}^{\text{exp}} = 0.953 \times \sigma_{\text{inc}}^{\text{exp}}$. The remaining 4.7% is possibly attributed to a charge-exchange or a two-step process: ^{36}S core excitation by inelastic scattering, followed by a proton removal. This is proposed based on their shifted parallel momentum distributions, their decay modes to negative parity states, their high excitation energy (around 4.7 MeV), and the fact that they were not observed in the $^{36}\text{S}(d, ^3\text{He})$ reaction study ([1985Kh04](#)).

E(level) [†]	J^π @	L [#]	C ² S&	Comments
0.0	1/2 ⁺	0	2.2 7	$b_f=31.4\%$ 72, $\sigma_f^{\text{sp}}=13.5$ mb, $R_s=0.55$ 11.
2388 1	3/2 ⁺	2	0.7 3	$b_f=7.8\%$ 36, $\sigma_f^{\text{sp}}=10.2$ mb, $R_s=0.52$ 10.
3862 1	5/2 ⁺	2	2.7 8	$b_f=27.3\%$ 58, $\sigma_f^{\text{sp}}=10.6$ mb, $R_s=0.49$ 10.
4101 2	(7/2 ⁻) [‡]			$b_f=-0.7\%$ 7.
4383 3	(5/2 ⁻) [‡]			$b_f=2.0\%$ 4.
4494 2	(7/2 ⁻)	(3)	0.30 7	L: 3 reported in 2016Mu03 . The parentheses are added by evaluators based on the statement in 2016Mu03 that the parallel momentum distribution has limited statistics and seems more consistent with L=3. $b_f=2.9\%$ 5, $\sigma_f^{\text{sp}}=10.7$ mb, $R_s=0.48$ 10. $b_f=9.5\%$ 10, $\sigma_f^{\text{sp}}=10.3$ mb, $R_s=0.47$ 9.
4667 2	5/2 ⁺	2	1.0 2	$b_f=0.9\%$ 3.
4768 2	(9/2 ⁻) [‡]			$b_f=0.3\%$ 2.
4962 3	(9/2 ⁻) [‡]			$b_f=1.5\%$ 2.
5089 3	(11/2 ⁻) [‡]			$b_f=14.5\%$ 14, $\sigma_f^{\text{sp}}=10.2$ mb, $R_s=0.47$ 9.
5200 2	5/2 ⁺	2	1.5 3	$b_f=1.4\%$ 10, $\sigma_f^{\text{sp}}=10.8$ mb, $R_s=0.47$ 9.
5710 2	(1/2 ⁻)	(1)	0.21 16	J^π : Interpreted as the deeply bound 1p _{1/2} proton removal. L: 1 reported in 2016Mu03 . The parentheses are added by evaluators based on the parallel momentum distribution (Fig. 3) in 2016Mu03 . $b_f=1.2\%$ 3, $\sigma_f^{\text{sp}}=10.2$ mb, $R_s=0.44$ 9. J^π : Interpreted as the deeply bound 1p _{1/2} proton removal. L: 1 reported in 2016Mu03 . The parentheses are added by evaluators based on the parallel momentum distribution (Fig. 3) in 2016Mu03 .

[†] From a least-squares fit to γ -ray energies.

[‡] Populated by a two-step reaction mechanism. [2016Mu03](#) states that they are likely to have $J \geq 5/2$ and negative parities.

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2016Mu03 deduced L by comparing the measured and eikonal-calculated parallel momentum distributions of residuals.

@ From the Adopted Levels, also assumed by 2016Mu03 for deducing C^2S .

& The spectroscopic factor $\text{C}^2\text{S} = b_f \sigma_{\text{inc}}^{\text{exp}} / R_s \sigma_f^{\text{sp}}$, where b_f is the population fraction, $\sigma_{\text{inc}}^{\text{exp}}$ is the measured total inclusive cross section, R_s is the quenching factor from a systematics study 2014To14, and σ_f^{sp} is the eikonal model calculated single-particle cross section.

 $\gamma(^{35}\text{P})$

E_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	E_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π
127 2	5089	(11/2 ⁻)	4962	(9/2 ⁻)	1473 2	3862	5/2 ⁺	2388	3/2 ⁺
237 2	4101	(7/2 ⁻)	3862	5/2 ⁺	1995 2	4383	(5/2 ⁻)	2388	3/2 ⁺
274 2	4768	(9/2 ⁻)	4494	(7/2 ⁻)	2279 2	4667	5/2 ⁺	2388	3/2 ⁺
321 2	5089	(11/2 ⁻)	4768	(9/2 ⁻)	2386 2	2388	3/2 ⁺	0.0	1/2 ⁺
391 2	4494	(7/2 ⁻)	4101	(7/2 ⁻)	2811 2	5200	5/2 ⁺	2388	3/2 ⁺
469 2	4962	(9/2 ⁻)	4494	(7/2 ⁻)	3860 2	3862	5/2 ⁺	0.0	1/2 ⁺
634 2	4494	(7/2 ⁻)	3862	5/2 ⁺	4668 2	4667	5/2 ⁺	0.0	1/2 ⁺
666 2	4768	(9/2 ⁻)	4101	(7/2 ⁻)	5202 2	5200	5/2 ⁺	0.0	1/2 ⁺
804 2	4667	5/2 ⁺	3862	5/2 ⁺	5709 2	5710	(1/2 ⁻)	0.0	1/2 ⁺
1337 2	5200	5/2 ⁺	3862	5/2 ⁺	7526 2	7527	(1/2 ⁻)	0.0	1/2 ⁺

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Level Scheme

